CSE 902: Homework 1

Due Date: 21 October 2026 | Total Points: 100

Dataset: Fashion-MNIST
Framework: PyTorch or TensorFlow

Fashion-MNIST contains 70,000 grayscale 28×28 images from 10 clothing categories: 60,000 training images and 10,000 test images. It can be loaded directly using the Torchvision Fashion-MNIST loader or the corresponding TensorFlow/Keras utility.

Learning Objectives

After completing this assignment, you should be able to:

1. Derive and implement the VAE objective and reparameterization estimator.
2. Distinguish reconstruction, prior sampling, interpolation, and latent traversal.
3. Explain how the KL term shapes a sampleable latent space.
4. Diagnose posterior collapse using multiple forms of evidence.
5. Evaluate how KL weighting and warm-up affect reconstruction, sampling, and latent utilization.

AI-Use and Academic-Integrity Policy

Generative-AI tools may be used for debugging, code suggestions, and conceptual clarification. You remain responsible for every submitted statement, result, and line of code. Submit an appendix identifying the tool, prompts used, substantive outputs incorporated, and your modifications. Undisclosed AI-generated work is an academic-integrity violation.
You must be able to reproduce, modify, and explain your work. The instructor may conduct a brief oral or live verification in which you explain a figure, identify an important line of code, predict the effect of changing β, or diagnose posterior collapse from your results.
By the way, you are more than welcome to discuss the homework with others in the class. But please ensure that the final solutions are developed by you!

General Experimental Requirements

Use the student-specific seed

$$s = 1000 + \text{the last three digits of your student ID.}$$

Use this seed for parameter initialization, data-loader shuffling, latent sampling, and visualization. State it in your report. Use the official training split for optimization and the official test split only for evaluation.

For every controlled comparison, state your prediction *before* presenting the result, then compare the observation with the prediction. An incorrect prediction followed by a technically sound analysis can receive full interpretation credit.

Required Implementation Starting Point

This assignment extends an established VAE implementation. You may reuse boilerplate for data download, device selection, checkpointing, and image saving, but you must implement the VAE-specific components identified below.

Reference implementations

- **Fashion-MNIST:** dataset repository
- **Torchvision loader:** official documentation
- **PyTorch VAE example:** official reference implementation
- **Keras VAE example:** official Keras reference implementation

The published examples are starting points, not solutions to copy unchanged. They use different architectures or datasets. Your implementation must follow the specifications below.

Data preprocessing

Convert each image to a tensor in $[0, 1]$. Do not normalize it to $[-1, 1]$, because the prescribed decoder uses a sigmoid output and binary cross-entropy reconstruction term.

Prescribed VAE architecture

Use latent dimension $d_z = 16$ and the convolutional encoder

$$1 \times 28 \times 28 \xrightarrow[\text{stride } 2]{\text{Conv } 1\to 32} 32 \times 14 \times 14 \xrightarrow[\text{stride } 2]{\text{Conv } 32\to 64} 64 \times 7 \times 7.$$

Use 4×4 kernels and padding 1. Flatten the encoder output and use separate linear layers for $\mu \in \mathbb{R}^{16}$ and $\log \sigma^2 \in \mathbb{R}^{16}$.

The decoder uses a linear projection to $64 \times 7 \times 7$, followed by

$$64 \times 7 \times 7 \xrightarrow{\text{ConvTranspose } 64\to 32} 32 \times 14 \times 14 \xrightarrow{\text{ConvTranspose } 32\to 1} 1 \times 28 \times 28.$$

Use 4×4 kernels, stride 2, and padding 1. Use ReLU for hidden activations and sigmoid at the image output.

Batch size	128
Optimizer	Adam
Learning rate	10^{-3}
Epochs	20
Reconstruction term	Binary cross-entropy, summed over pixels
Latent dimension	16
Prior	$p(z) = \mathcal{N}(0, I)$

What you must implement

You must implement or complete: the encoder heads for μ and $\log \sigma^2$; reparameterization; the β-VAE objective; prior sampling; linear interpolation; per-dimension KL diagnostics; active-unit counting; latent replacement and shuffling tests; KL warm-up; and interpretation of all results. You may reuse general training-loop and plotting patterns.

Problem 1: VAE Foundations and Implementation (20 points)

Consider

$$q_\phi(z \mid x) = \mathcal{N}\Big(\mu_\phi(x), \mathrm{diag}(\sigma^2_\phi(x))\Big), \qquad p_\theta(x \mid z), \qquad p(z) = \mathcal{N}(0, I).$$

A. ELBO and loss function (8 points)

Starting with $\log p_\theta(x)$, explain why direct marginal-likelihood optimization is difficult and state the evidence lower bound (ELBO). Show how minimizing the negative ELBO leads to

$$\mathcal{L}_\beta(x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] + \beta D_{\mathrm{KL}}\big(q_\phi(z \mid x) \| p(z)\big).$$

For the diagonal Gaussian posterior, write the closed-form KL term used in your implementation. Explain the distinct roles of reconstruction and KL.

B. Reparameterization and implementation check (6 points)

Implement

$$\epsilon \sim \mathcal{N}(0, I), \qquad z = \mu + \sigma \odot \epsilon, \qquad \sigma = \exp\big(\tfrac{1}{2}\log \sigma^2\big).$$

Explain why sampling z directly from $q_\phi(z \mid x)$ obstructs ordinary backpropagation and how reparameterization resolves the issue. Verify tensor shapes at every encoder and decoder stage and report the number of trainable parameters.

C. Reconstruction versus generation (6 points)

Explain the difference between (i) reconstructing an observed image using a code from $q_\phi(z \mid x)$ and (ii) generating a new image using $z \sim \mathcal{N}(0, I)$. Why can a model reconstruct well while generating poor prior samples? Contrast the VAE with a deterministic autoencoder in this respect.

Deliverables. Derivation; closed-form KL; reparameterization code; shape/parameter table; concise conceptual answers.

Problem 2: KL Weighting and Latent-Space Geometry (30 points)

Train the prescribed architecture using all four configurations:

Configuration	β	Schedule
VAE-A	0	Constant
VAE-B	1	Constant
VAE-C	4	Constant
VAE-D	4	Linear warm-up from 0 to 4 during epochs 1–10

A. Training, reconstruction, and prior sampling (12 points)

For every configuration, plot training and test reconstruction loss, KL divergence, and total objective by epoch. Display test reconstructions as paired original/reconstruction grids. Generate 64 samples from the same fixed prior codes for all applicable comparisons. Report both the final metrics and the best test reconstruction epoch.
Interpret the reconstruction–regularization tradeoff. Do not assume that larger β must improve sample quality; state what your evidence shows.

B. Aggregate latent distribution (8 points)

Encode the complete test set and visualize the latent means using PCA to two dimensions, colored by Fashion-MNIST class. Using the same fitted PCA transformation, overlay or separately plot an equal number of samples from $\mathcal{N}(0, I)$. Discuss class overlap, gaps, isolated regions, scale mismatch, and excessive concentration.

Explain why a two-dimensional projection cannot establish prior matching or semantic disentanglement in the original 16-dimensional space.

C. Interpolation and traversal (10 points)

Choose two test images from different classes. Use their posterior means as endpoints and decode at least 11 linear interpolation points:

$$z(\alpha) = (1 - \alpha)z_a + \alpha z_b, \qquad \alpha \in [0, 1].$$

Repeat for VAE-A, VAE-B, VAE-C, and VAE-D. In addition, for VAE-B, select the three dimensions with the largest $\mathrm{Var}_x[\mu_j(x)]$ and traverse each one over $\{-3, -2, -1, 0, 1, 2, 3\}$ while holding all other coordinates fixed.

Discuss continuity, plausibility, and coupled attribute changes. Explain why a smooth interpolation does not prove that the latent variables are disentangled.

Deliverables. Curves; reconstruction and prior-sample grids; PCA plots; four interpolation sequences; three traversal grids; interpretation.

Problem 3: Diagnosing Posterior Collapse (30 points)

A. Per-dimension diagnostics (12 points)

For each latent dimension j, compute

$$\mathrm{KL}_j = \frac{1}{2N} \sum_{i=1}^{N} \left(\mu_{ij}^2 + \exp(\log \sigma_{ij}^2) - \log \sigma_{ij}^2 - 1 \right).$$

Call a dimension active when $\mathrm{KL}_j > 0.01$. For each model, report:

- all 16 values of KL_j and their sum;
- the number of active dimensions;
- $\mathrm{Var}_x[\mu_j(x)]$ for every dimension; and
- a bar plot of per-dimension KL values with the 0.01 threshold shown.

Explain why a small total KL may mean either successful regularization or failure to encode input-dependent information.

B. Interventions on the latent code (8 points)

For a fixed batch of at least 64 test examples, compare ordinary reconstructions with outputs obtained after:

1. replacing every posterior code by an independent prior sample; and
2. independently permuting the posterior codes across examples.

Quantify the mean absolute output change and show representative grids. If the decoder is using z, what should each intervention do? Explain why the interventions provide evidence beyond the KL value alone.

C. Does KL warm-up help? (10 points)

Compare VAE-C with VAE-D. Determine whether warm-up improves latent utilization without unacceptable reconstruction degradation. Base your conclusion on active units, per-dimension KL, latent-intervention sensitivity, reconstruction, and prior samples—not on one metric alone.

Complete collapse is not guaranteed with this decoder. If no model collapses completely, identify the configuration showing the strongest movement toward collapse and justify the claim. You may test free bits, cyclical scheduling, or a weaker decoder for up to 3 bonus points.

Deliverables. KL plots and table; intervention grids and numerical sensitivity; VAE-C versus VAE-D evidence table; conclusion.

Problem 4: Evaluating Latent-Space Usefulness (20 points)

A. A multi-criterion comparison (10 points)

Create a single table comparing VAE-A through VAE-D on:

- reconstruction quality;
- prior-sample plausibility and diversity;
- aggregate-posterior compatibility with the prior;
- number of active dimensions;
- sensitivity to latent replacement and shuffling; and
- interpolation/traversal behavior.

Select the most useful model for each of these two purposes: (i) compression and reconstruction, and (ii) generation from the prior. The same model need not win both. Define the evidence used for each selection.

B. Final technical interpretation (10 points)

Answer each question in a short, evidence-based paragraph:

1. Which configuration uses z most strongly, and how do you know?
2. Which configuration produces the most defensible prior samples?
3. Is the model with the smoothest interpolation necessarily the best VAE?
4. What is your strongest evidence for or against posterior collapse?
5. If two VAEs reconstruct equally well, what additional evidence determines which learned the more useful latent space?

Deliverables. Comparison table and five concise conclusions tied to specific figures or metrics.

Submission Requirements

Submit:

1. A PDF report of at most 10 pages, excluding references and AI-use appendix.
2. Runnable source code and a README with exact reproduction commands.
3. Configuration files, checkpoints, fixed prior codes, and raw metric files.
4. One unsuccessful or unexpected result and a technically grounded diagnosis.
5. An AI-use appendix containing prompts, outputs used, and your modifications.

Every figure must state the configuration, seed, and conclusion it supports. A gallery of visually appealing samples without quantitative or diagnostic analysis will receive limited credit.

Grading

Component	Points
Problem 1: VAE foundations and implementation	20
Problem 2: KL weighting and latent-space geometry	30
Problem 3: Diagnosing posterior collapse	30
Problem 4: Evaluating latent-space usefulness	20
Total	**100**

Grading emphasizes technical correctness, controlled comparisons, and reproducibility. It also emphasizes diagnosis using multiple forms of evidence, interpretation of unexpected results, and the ability to defend the submitted work. A carefully diagnosed failure can earn more credit than an apparently successful model supported only by visually pleasing images.